

# DUAL-ATTENTION AUTOENCODER FOR REAL-TIME TRANSACTION FRAUD DETECTION

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## ABSTRACT:

Payment fraud detection systems must balance high true-positive detection against catastrophic false-positive customer disruptions. Extreme class imbalance severely undermines conventional supervised classifiers. In this paper, we propose Dual-Attention Autoencoder DA-AE, an unsupervised reconstruction and latent attention mode

## 1. RESEARCH PROBLEM:

Severe class imbalance <0.2% fraud and adversarial feature shift in credit transaction str

## 2. OBJECTIVE:

Construct an unsupervised dual-attention autoencoder that reconstructs normal transaction manifolds and flags anomalous reconstruction errors in real-time <15ms.

## 3. DATASET:

Credit Card Fraud benchmark comprising 284,807 transactions. Features: PCA latent vectors transaction Amount, and timestamp.

## 4. METHODOLOGY & ARCHITECTURE:

- RobustScaler applied to Amount and Time to mitigate outliers.
- 4-layer Encoder 29 -> 64 -> 32 -> 16 with multi-head latent attention weighting.
- Symmetric Decoder and Mean Squared Reconstruction Error thresholding.

## 5. EVALUATION:

AUPRC: 0.884, F1-Score: 0.842, p99 Latency: <12ms.